Quantum circuits

Quantum computing

G. Chênevert

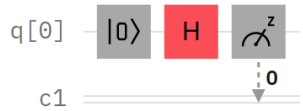
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A hands-on approach

This is what the "Hello world" program looks like in quantum computing:



Let's create accounts and log in to

IBM Quantum Platform

https://quantum.cloud.ibm.com/

in order to run it on a small quantum computer (and/or simulator)

Quantum computing

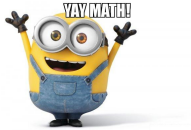
The basic unit of information in classical computing is a **bit**

that can take two symbolic values: 0 and 1

Quantum computing works with **quantum bits** (or **qubits**)

that can be simultaneously 0 and 1 — in various proportions!

To make sense of what it means, we first need to review some concepts and formalism from Quantum Mechanics



Quantum circuits

Quantum bits

Qubit states

Quantum gates

Multiple qubit gates

Let's ignore all the physics

A **quantum bit** is a physical system for which the (stationary) states can be described as nonzero *complex* linear combinations

$$|\varphi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle.$$

Classical states:

- $\alpha = 1,\ \beta = 0 :\ |\varphi\rangle = |0\rangle$
- $\alpha = 0,\ \beta = 1 :\ |\varphi\rangle = |1\rangle$

Dirac formalism

Concretely: the state of a qubit is just a 2-dimensional vector

$$|\varphi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \begin{bmatrix}\alpha\\ \beta\end{bmatrix} \in \mathcal{V}_2$$

expressed in a basis

$$|0\rangle = \begin{bmatrix}1\\ 0\end{bmatrix} \qquad \text{and} \qquad |1\rangle = \begin{bmatrix}0\\ 1\end{bmatrix}$$

of orthonormal vectors for the standard hermitian product

$$\langle\varphi|\psi\rangle = \alpha^*\,\gamma + \beta^*\,\delta = \underbrace{\begin{bmatrix}\alpha^* & \beta^*\end{bmatrix}}_{|\varphi\rangle^\dagger} \underbrace{\begin{bmatrix}\gamma\\ \delta\end{bmatrix}}_{|\psi\rangle}$$

Measurement

The state of a qubit can be thought of as a nonzero linear combination

$$|\phi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \qquad \alpha, \beta \in \mathbb{C}.$$

When we measure it, the classical outcomes $|0\rangle$ and $|1\rangle$ are observed with probabilities

$$\mathbb{P}\big[\,\mathcal{M}|\phi\rangle = |0\rangle\,\big] = \frac{|\alpha|^2}{|\alpha|^2 + |\beta|^2}, \qquad \mathbb{P}\big[\,\mathcal{M}|\phi\rangle = |1\rangle\,\big] = \frac{|\beta|^2}{|\alpha|^2 + |\beta|^2}.$$

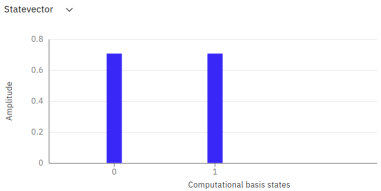
In general, the probability of a basic state $|b\rangle$ being observed is

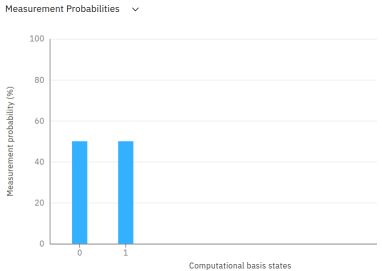
$$\mathbb{P}\big[\,\mathcal{M}|\phi\rangle = |b\rangle\,\big] = \frac{|\langle\varphi|b\rangle|^2}{\|\varphi\|^2}.$$

Example

What are the possible outcomes of the measurement of the qubit state $|\varphi\rangle = |0\rangle + |1\rangle$?

$$\mathbb{P}\big[\,\mathcal{M}|\varphi\rangle = |0\rangle\,\big] = \mathbb{P}\big[\,\mathcal{M}|\varphi\rangle = |1\rangle\,\big] = \frac{1}{2}$$





Quantum circuits

Quantum bits

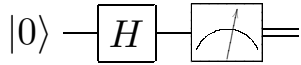
Qubit states

Quantum gates

Multiple qubit gates

Hello World revisited

Ok so we implemented our first quantum program: a True Random Bit Generator



where H is the **Hadamard gate** that turns $|0\rangle$ into $\frac{|0\rangle + |1\rangle}{\sqrt{2}}$



Equivalent states

Remark: if $|\varphi\rangle$ and $|\varphi'\rangle$ are two globally proportional states:

$$|\varphi\rangle = \gamma\,|\varphi'\rangle \qquad \text{with } \gamma \in \mathbb{C} \setminus \{0\}$$

then $|\varphi\rangle$ and $|\varphi'\rangle$ cannot be distinguished by measurements:

$$\mathbb{P}\big[\,\mathcal{M}|\varphi'\rangle = |b\rangle\,\big] = \frac{|\langle\varphi'|b\rangle|^2}{\|\varphi'\|} = \frac{|\gamma|^2\,\langle\varphi|b\rangle}{|\gamma|^2\,\|\varphi\|^2} = \frac{\langle\varphi|b\rangle}{\|\varphi\|^2} = \mathbb{P}\big[\,\mathcal{M}|\varphi\rangle = |b\rangle\,\big].$$

We write $|\varphi\rangle \sim |\varphi'\rangle$.

Equivalence and normalization

Quantum states should really be thought of as *equivalence classes of vectors*

$$\{\, \alpha\,|\phi\rangle \;\mid\; \gamma \neq 0 \,\} \qquad \textit{i.e.}\ \text{lines in } \mathcal{V}_2$$

Clearly any quantum state is equivalent to a normalized state

$$|\phi\rangle \;\sim\; \frac{1}{\|\phi\|}\,|\phi\rangle$$

but such a normalized state is *not* unique:

$$|\phi\rangle \;\sim\; \gamma\,|\phi\rangle,$$

another state with the same norm, iff $|\gamma| = 1$, *i.e.* $\gamma = e^{it}$ $(t \in \mathbb{R})$

Visualizing qubit states

So consider a qubit in quantum state

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle.$$

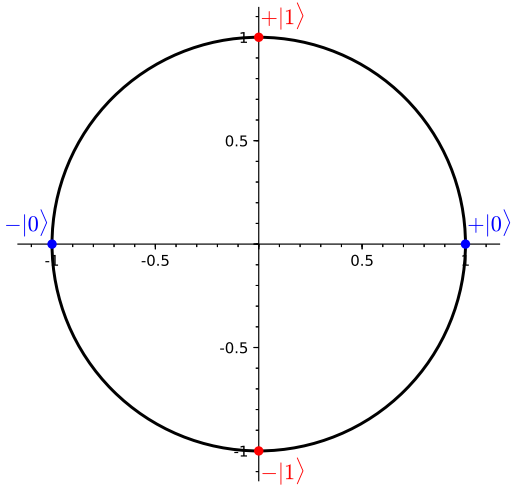
In general α and β are complex numbers: hard to visualize! ($\dim_{\mathbb{R}} \mathcal{V}_2 = 4$)

Let us assume for the moment that $\alpha, \beta \in \mathbb{R}$.

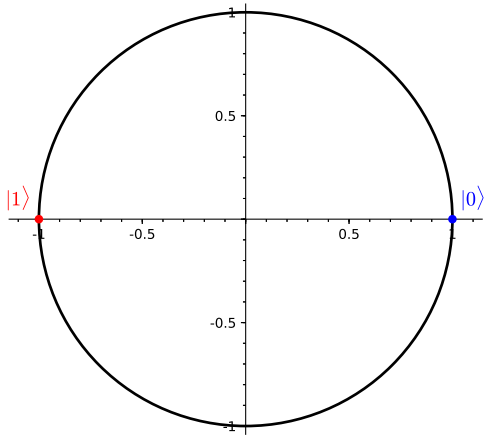
Since $|\psi\rangle \sim \dfrac{1}{\|\psi\|}|\psi\rangle$, we can assume without loss of generality that $\alpha^2 + \beta^2 = 1$.

Looks like a circle...

A circle ?



Yes: the Bloch circle



The (real) Bloch representation

According to the first picture we are tempted to write:

$$|\psi\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle \qquad 0 \leq \theta < 2\pi$$

but this representation has the ambiguity $\theta \longleftrightarrow \theta + \pi$, $|\psi\rangle \sim -|\psi\rangle$.

In the second, more accurate picture, what we actually see is the point

$$P_{|\psi\rangle} = (\cos 2\theta,\, \sin 2\theta).$$

Thus in hindsight it would have been better to write, non-ambiguously,

$$|\psi\rangle = \cos(\tfrac{\theta}{2})\,|0\rangle + \sin(\tfrac{\theta}{2})\,|1\rangle.$$

Angle between two states

In the (real) Bloch representation:

$$\begin{cases} |\psi\rangle = \cos(\frac{\theta_1}{2})\,|0\rangle + \sin(\frac{\theta_1}{2})\,|1\rangle, \\ |\phi\rangle = \cos(\frac{\theta_2}{2})\,|0\rangle + \sin(\frac{\theta_2}{2})\,|1\rangle \end{cases}$$

we have

$$\langle\phi\,|\,\psi\rangle = \cos(\tfrac{\theta_1}{2})\cos(\tfrac{\theta_2}{2}) + \sin(\tfrac{\theta_1}{2})\sin(\tfrac{\theta_2}{2}) = \cos\tfrac{\theta_1-\theta_2}{2}.$$

In particular:

$$\langle\phi\,|\,\psi\rangle = 0 \iff \tfrac{\theta_1-\theta_2}{2} = \pm\tfrac{\pi}{2} \iff \theta_2 = \theta_1 \pm \pi.$$

Orthogonal states lie *opposite* on the Bloch circle.

Exercise

Locate the following normalized states on the Bloch circle:

$$|\varphi_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|\varphi_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$|\varphi_3\rangle = \frac{-|0\rangle + \sqrt{3}\,|1\rangle}{2}$$

$$|\varphi_3\rangle = 0.6\,|0\rangle + 0.8\,|1\rangle$$

Towards the general Bloch representation

Now for a general state $|\psi\rangle \neq 0$:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle, \qquad \alpha, \beta \in \mathbb{C}.$$

Without loss of generality we can assume $|\alpha|^2 + |\beta|^2 = 1$ (normalized state).

Equivalent normalized states: if $|\psi\rangle \sim |\phi\rangle$, then $|\psi\rangle = \gamma\,|\phi\rangle$ with $|\gamma| = 1$.

So: if $\alpha = A\,e^{ia}$, by multiplying by e^{-ia} we can reduce to the case

$$\alpha = A \text{ is real}, \quad \beta = B\,e^{ib}, \quad A^2 + B^2 = 1.$$

Bloch representation

$$|\psi\rangle \sim A\,|0\rangle + B\,e^{ib}\,|1\rangle, \quad A^2 + B^2 = 1.$$

From the real case we know we should write $A = \cos(\frac{\theta}{2})$, $B = \sin(\frac{\theta}{2})$.

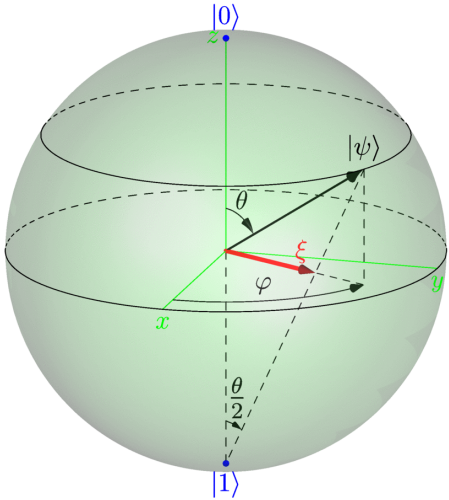
We have proved:

Every qubit state is equivalent to a unique normalized state of the form

$$\cos(\tfrac{\theta}{2})\,|0\rangle + \sin(\tfrac{\theta}{2})\,e^{i\varphi}\,|1\rangle.$$

These correspond to points $(\cos\varphi\,\sin\theta,\ \sin\varphi\,\sin\theta,\ \cos\theta)$ on a *sphere*.

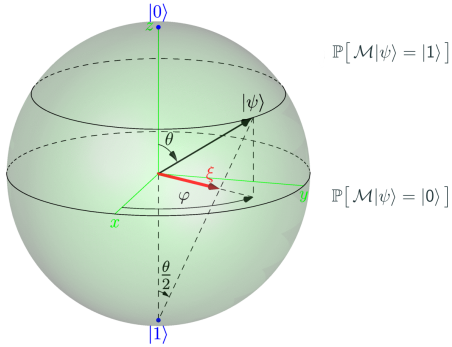
The Bloch sphere $\mathcal{B}$ (click title for interactive model)



http://stla.github.io/stlapblog/posts/BlochSphere.html

Properties of the Bloch representation

- Pairs of orthogonal states correspond to antipodal points on the Bloch sphere.
- The probability that $|\psi\rangle$ is measured as $|0\rangle$ or $|1\rangle$ can be interpreted as relative areas on the sphere.



Exercise

Locate equivalent states to the following qubit states on the Bloch sphere:

a) $|0\rangle$

b) $|1\rangle$

c) $|0\rangle + |1\rangle$

d) $\dfrac{|0\rangle + i\,|1\rangle}{\sqrt{2}}$

e) $3\,|0\rangle - 4i|1\rangle$

Quantum circuits

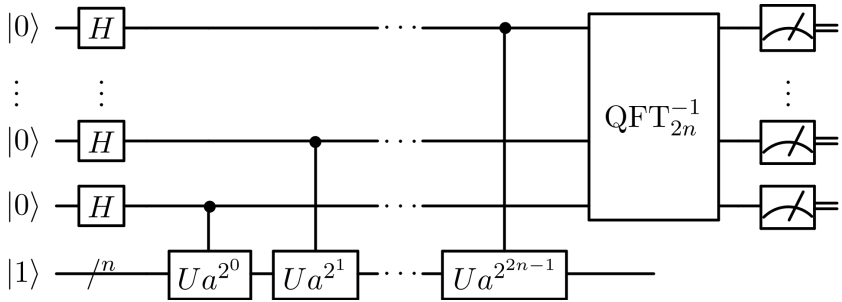
Quantum bits

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Multiple qubit gates

Spoilers ahead: Shor's algorithm



Quantum circuits

Quantum circuits are made up of

- **quantum registers** containing qubits
- **quantum logic gates** modifiying the state of these qubits
- **classical registers** containing regular bits
- **measurements** mapping quantum registers to classical registers

that can then be manipulated with a classical electronic circuit.

NOT gate

$$\begin{cases} \mathsf{NOT}\,|0\rangle = |1\rangle \\ \mathsf{NOT}\,|1\rangle = |0\rangle \end{cases}$$

$$\mathsf{NOT}(\alpha\,|0\rangle + \beta\,|1\rangle) = \alpha\,|1\rangle + \beta\,|0\rangle$$

$$\mathsf{NOT}\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = \begin{bmatrix}\beta\\ \alpha\end{bmatrix} \qquad \mathsf{NOT} = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$$

Interpretation on the Bloch sphere

Fixed points of NOT:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

NOT can be thought of as a *rotation of π around the x-axis*

often called the **Pauli X** gate for this reason and written NOT, X or $\oplus$

Note: $X^2 = I$

Exercise

Geometrical interpretation of the quantum gates

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \qquad \text{and} \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \ ?$$

Phase gate $P = P(\theta)$

$$P = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}$$

$$P|0\rangle = |0\rangle, \qquad P|1\rangle = e^{i\theta}$$

$$P(\alpha\,|0\rangle + \beta\,|1\rangle) = \alpha\,|0\rangle + e^{i\theta}\,\beta\,|1\rangle$$

Remark : $Z := P(\pi) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ special case

Hadamard gate

$$H = \frac{X+Z}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Sends $|0\rangle$ to $H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$, $|1\rangle$ to the orthogonal state $H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$.

Remark: $H^2 = I$ (isn't it?)

Equivalent gates

Looking at the IBM Composer, we see that the H gate in our first program was transformed into

$$RZ\ \sqrt{X}\ RZ$$

Universal gate U

Depends on 3 parameters θ, φ and λ:

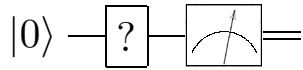
$$U = \begin{bmatrix} \cos(\frac{\theta}{2}) & -e^{i\lambda}\sin(\frac{\theta}{2}) \\ e^{i\varphi}\sin(\frac{\theta}{2}) & e^{i(\lambda+\varphi)}\cos(\frac{\theta}{2}) \end{bmatrix}$$

All gates encountered so far are special cases !

Remark: U is a unitary matrix ($U^\dagger U = I$) and all previous gates are a special case

Exercise: general TRBG

Write a quantum program that outputs $|1\rangle$ with probability 62%.



Solution: it suffices to use a universal gate U with $\theta \approx 1.81316$ (why??)

You may verify on IBM Q that it actually works.

General single-qubit gate

Theorem

The time evolution operator on the space of stationary states of a quantum system is represented by a unitary matrix.

This means that any physically realizable action G (that doesn't destroy superposition) on a qubit preserve orthogonality between states:

$$\langle G\varphi|G\psi\rangle = |G\varphi\rangle^\dagger|G\psi\rangle = (|\varphi\rangle^\dagger G^\dagger)(G|\psi\rangle) = |\varphi\rangle^\dagger(G^\dagger G)|\psi\rangle = |\varphi\rangle^\dagger|\psi\rangle = \langle\varphi|\psi\rangle.$$

The images $G|0\rangle$ and $G|1\rangle$ are thus an orthonormal basis and we can show that there exists parameters θ, φ and λ such that $G = U(\theta, \varphi, \lambda)$.

Other point of view (geometrical interpretation)

Any 2×2 unitary matrix G admits an orthogonal eigenbasis $|\psi_0\rangle$, $|\psi_1\rangle$ for which

$$\begin{cases} G\,|\psi_0\rangle = e^{+i\sigma}\,|\psi_0\rangle \\ G\,|\psi_1\rangle = e^{-i\sigma}\,|\psi_1\rangle \end{cases}$$

If Q denotes the unitary transformation for which $Q\,|0\rangle = |\psi_0\rangle$ and $Q\,|1\rangle = |\psi_1\rangle$, then

$$Q^\dagger G Q = \begin{bmatrix} e^{+i\sigma} & 0 \\ 0 & e^{-i\sigma} \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & -2\sigma \end{bmatrix} = P(-2\sigma).$$

On the Bloch sphere, *G is a rotation of angle -2σ around the axis through the orthogonal states $|\psi_0\rangle$ and $|\psi_1\rangle$* .

Universal family

Remark: every single qubit gate G can be expressed as a combination of

$$H \qquad \text{and} \qquad P(\theta) \qquad (\theta \in \mathbb{R}) \qquad \text{only.}$$

Idea:

- express G as a combination of $R_x(\alpha)$, $R_y(\beta)$, $R_z(\gamma)$
- explicit formulas for these 3 kinds of rotations

Corollary: every single qubit gate G can be *approximated* by a combination of

$$H \qquad \text{and} \qquad P(\tfrac{2\pi}{n}) \qquad (n \gg 0) \qquad \text{only.}$$

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N-level quantum systems

We may work in general with quantum systems for which the vector space of stationary states $\mathcal{V}_N$ is N-dimensional.

Any state $|\varphi\rangle \in \mathcal{V}_n$ can be written uniquely as a complex linear combination

$$|\varphi\rangle = \sum_{n<N} \alpha_n |n\rangle, \qquad \alpha_n \in \mathbb{C}$$

where $|0\rangle$, $|1\rangle$, $\ldots$, $|N-1\rangle$ is an orthonormal family called the **computational basis**

Special case $N = 2$: this is just our description of qubits

(What about $N = 1$?)

2-qubit system

Consider a system with two qubits A and B. Suppose:

A is in state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$

B is in state $|\phi\rangle = \gamma\,|0\rangle + \delta\,|1\rangle$

Then the system (A, B) is in state

$$|\psi\rangle \otimes |\phi\rangle = (\alpha\,|0\rangle + \beta\,|1\rangle) \otimes (\gamma\,|0\rangle + \delta\,|1\rangle)$$

$$= \alpha\gamma\,|0\rangle \otimes |0\rangle + \alpha\delta\,|0\rangle \otimes |1\rangle + \beta\gamma\,|1\rangle \otimes |0\rangle + \beta\delta\,|1\rangle \otimes |1\rangle$$

$$= \alpha\gamma\,|00\rangle + \alpha\delta\,|01\rangle + \beta\gamma\,|10\rangle + \beta\delta\,|11\rangle$$

2-qubit system

More generally: the 2-qubit system can be in *any* linear combination state

$$a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle \in \mathcal{V}_2 \otimes \mathcal{V}_2 = \mathcal{V}_4$$

Some of these can*not* be written in the form $|\psi\rangle \otimes |\phi\rangle$: called **entangled**

Example

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}} \qquad \textbf{Bell state}$$

Two qubit gates

Do we have the analogues of the classical AND, OR, XOR, NAND, ... gates for quantum bits?

NO! They lose information...

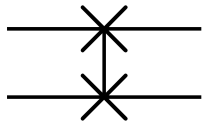
Recall: the space of quantum states for a system of 2 qubits is

$$\mathcal{V}_2 \otimes \mathcal{V}_2 = \mathcal{V}_4$$

basis $|0\rangle \otimes |0\rangle$, $|0\rangle \otimes |1\rangle$, $|1\rangle \otimes |0\rangle$, $|1\rangle \otimes |1\rangle$ or $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ or $|0\rangle$, $|1\rangle$, $|2\rangle$, $|3\rangle$

2-qubit gates are represented by 4×4 *unitary matrices*

SWAP gate



$$|\psi\rangle \otimes |\phi\rangle \;\mapsto\; |\phi\rangle \otimes |\psi\rangle$$

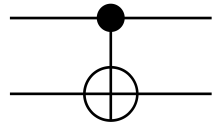
$$\mathsf{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \mathsf{diag}(1, X, 1)$$

CNOT = CX gate

$$\mathrm{CX}(|x\rangle \otimes |y\rangle) \text{ "} = \text{"} X^x|y\rangle = \begin{cases} |y\rangle & \text{if } |x\rangle = |0\rangle \\ X|y\rangle & \text{if } |x\rangle = |1\rangle \end{cases} = |x \oplus y\rangle$$

To be able to go back we must output $|x\rangle$ as well:

$$\text{"}\mathrm{CX} \begin{bmatrix} |x\rangle \\ |y\rangle \end{bmatrix} = \begin{bmatrix} |x\rangle \\ |x \oplus y\rangle \end{bmatrix} \text{"}$$



CNOT = CX gate

Proper way to write this:

$$\mathrm{CX}(|x\rangle \otimes |y\rangle) = |x\rangle \otimes (|x \oplus y\rangle)$$

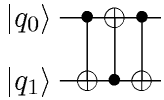
$$\mathrm{CX}\big(|0\rangle \otimes |\phi\rangle\big) = |0\rangle \otimes |\phi\rangle \qquad \mathrm{CX}\big(|1\rangle \otimes |\phi\rangle\big) = |1\rangle \otimes X|\phi\rangle$$

$$\mathrm{CX} = \mathrm{diag}(I, X) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Reversible operation ! $\mathrm{CX}^2 = I$

Exercise

What is the effect of 3 consecutive CNOT gates like below?



Answer: a SWAP gate

Exercise

What is the matrix representation of the 2-qubit gate corresponding to the application of X on the first qubit and H on the second qubit?

Answer:

$$X \otimes H = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$

This can be verified on IBM Q by applying $X \otimes H$ to $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

!! Watch out for bit-order conventions !!

Other boolean operations

So far we know how to build quantum circuits computing NOT and XOR.

What about other logical gates, *e.g.* AND and OR?

Recall that quantum gates are represented by unitary matrices, *hence always reversible*.

$\Longrightarrow$ ancillary qubits are needed

The Toffoli gate

Reversibly computes the logical AND of two qubits:

$$\text{Toff}(|x\rangle\,|y\rangle\,|z\rangle) = |x\rangle\,|y\rangle\,|z \oplus \text{AND}(x,y)\rangle$$

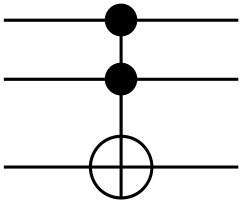
Note:

$$\text{Toff}(|x\rangle\,|y\rangle\,|z\rangle) = \begin{cases} |x\rangle\,|y\rangle\,|z\rangle & \text{if } x = 0 \\ |x\rangle\,|y\rangle\,|y \oplus z\rangle & \text{if } x = 1 \end{cases}$$

so the Toffoli gate may be viewed as a controlled-CNOT gate.

The Toffoli (or CCNOT) gate

$$\mathrm{Toff}(|x\rangle\,|y\rangle\,|z\rangle) = |x\rangle\,|y\rangle\,|z \oplus \mathrm{AND}(x,y)\rangle$$



$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & & 1 \\ & & & & & & 1 & \end{bmatrix}$$

Reversible evaluation

More generally: $f : \{0,1\}^n \to \{0,1\}$ is any boolean function

consider the gate U_f on $n+1$ qubits defined by

$$U_f : |\mathbf{x}\rangle \otimes |y\rangle \mapsto |\mathbf{x}\rangle \otimes |y \oplus f(\mathbf{x})\rangle.$$

You may check that this is a unitary operator on $\mathcal{V}_{2^n} \otimes \mathcal{V}_2 = (\mathcal{V}_2)^{\otimes(n+1)}$

that allows to evaluate f on $\mathbf{x}$ without losing $\mathbf{x}$:

$$U_f(|\mathbf{x}\rangle|0\rangle) = |x\rangle|f(\mathbf{x})\rangle.$$

Fact

For any boolean function $f : \{0,1\}^n \to \{0,1\}$

the quantum gate U_f can be implemented using only Toffoli and X gates.

Classical analogue: any boolean function can be generated using only the NAND gate

Exercise: come up with a circuit that implements the U_f gate for $f(x, y) = x \,\mathrm{OR}\, y$

Quantum gates, general case

General case of a n-qubit system:

$$\underbrace{\mathcal{V}_2 \otimes \cdots \otimes \mathcal{V}_2}_{n} \cong \mathcal{V}_{2^n}$$

Any reversible quantum operation can be viewed as a $2^n \times 2^n$ unitary matrix:

$$G \in \mathrm{U}(2^n).$$

Usually described as a **quantum circuit** made of gates on smaller numbers of qubits.